Structured Output and Schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Structured Output and Schemas

> Prepare Structured Output and Schemas: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Separate JSON parsing, **schema validation** and source support checks

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Prompt as a Contract

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt specification with observable response criteria

Reference example, not the only acceptable wording: Teacher explanation: **Field**s status,room,capacity,source_ids.

Status supported or needs_review. supported requires matching current approval and current capacity for the same room; no unresolved conflict.

A Distribution of Possibilities

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a next-token table is a valid probability distribution

Reference example, not the only acceptable wording: Teacher explanation: Entries are nonnegative and sum to 1. Not ready has mass 0.3+0.2 = 0.5, including busy and unknown.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> **Schema** requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

Concept to explain

Week 8: Structured **Output** and **Schema**s

Term	Meaning
schema	A specification of required fields, types and structural rules for data.
field	A named item of information in a record or structured message.
type	The category of a value, such as text, number or Boolean.
validation	Checking data or behaviour against rules; in ML, validation data is used to choose settings or models, separately from final testing.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

For the compact core route, use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

No. Parse and validate the returned value, then check its claims and permitted actions against the **evidence** and contract.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can validate structured **output** in layers: syntax, **schema** and source meaning.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Structured Output and Schemas using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Schema field**s: answer=text; **evidence**=list; confidence=low/medium/high.

2. **Output** omits **evidence**: **validation** fails.

3. Confidence=97 is the wrong **type** and allowed-value set.

4. Reject, repair, or request again before downstream use.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> **Schema** requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

> ### The teaching case
>
> Diagnose the shape error.

Exact explanation to model

> ### Why the result follows
>
> Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Explain the added worked example

Explain the added worked example · 1

A response may be valid JSON and still violate the **schema** or sources. First parse the text. Then check the exact object **field**s and **type**s. Finally check source support and domain meaning.

These checks answer different questions. Do not call a successful parser a complete validator.

Explain the added worked example · 2

Use the Week 11 closed object: exactly status,room,capacity,source_ids. status is supported or needs_review. source_ids is a list of distinct known source-ID strings.

For supported, room is a nonempty string and capacity a nonnegative integer. For needs_review, room and capacity are null. Boolean values are not capacities under this contract.

Explain the added worked example · 3

These fragments illustrate failures, not complete candidate objects. A real check receives the whole object and verifies every required **field**.

Do not repair a string capacity automatically unless a separate documented conversion policy permits it; otherwise report **validation** failure. A null value differs from missing a **field**.

Key terms

Term	Meaning
Syntax validation	Checking whether text can be parsed under the format grammar
Schema validation	Checking fields, types and allowed combinations in a parsed value
Semantic validation	Checking meaning and consistency with sources or domain rules
Closed schema	A contract rejecting fields beyond the stated set

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

Suppose sources P and Q approve room B and give capacity 12.

The complete object {"status":"supported","room":"B","capacity":99,"source_ids":["P","Q"]} parses and satisfies the stated **field type**s, but its capacity is unsupported by Q. Reject it at **semantic validation**.

Listing a real source ID does not prove that source supports the associated claim.

Model the reasoning aloud · 5

An object with needs_review and room B violates the conditional **schema** rule even if its source IDs are real. An object with supported, room B and capacity 12 but source ID Z fails the known-source rule.

Require the linked current approval and capacity records to support the same room.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Structured Output and Schemas

A fictional Study Buddy returns a machine-readable book fact.

The facts we will use

Supplied facts

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

Supplied facts

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

AI ACADEMY / AI-3/8

Structured Output and Schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Parse

Each shown object is valid JSON syntax.



2. Schema check

Y fails because its pages value is text, not an integer.



3. Meaning check

Z fits field types but contradicts R's page count.



4. Accept X

X passes syntax, declared schema and source comparison.

CHANGE ONE THING

Add an extra field "rating":5 to X under the exactly-fields schema.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Parse	Each shown object is valid JSON syntax.
2. Schema check	Y fails because its pages value is text, not an integer.
3. Meaning check	Z fits field types but contradicts R's page count.
4. Accept X	X passes syntax, declared schema and source comparison.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

What this example does not establish

Keep the stated limit

Source agreement does not establish that the approved catalogue itself is correct or complete.

Change one thing in the pictured case

Change one thing

Change: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Predicted result

Prediction: The changed object fails **schema** despite keeping the correct page count.

Why the result follows

Reason: Exactly means additional **field**s are not allowed in this contract.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Structured Output and Schemas, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: **Schema** requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

2. State the decision to explain

Write the question to explain: Diagnose the shape error.

3. Reveal the worked reasoning

Reveal after **prediction**s: Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Hold back the independent task

Keep for an independent attempt: A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: D/8 with P/Q passes syntax, **schema** and source consistency. Zero is a permitted integer but is supported only if Q says zero. Exit: **schema** must reject Boolean capacity and extra **field**s; no action authority exists.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read the explanation in “Notice the evidence” and “See how it works”, then complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the “A closer explanation” material in “See how it works” and “Follow a worked example”, then complete W2. The pictorial case is an additional example.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Reasoning to listen for

Teacher check: Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: JSON syntax says nothing about **schema** or source truth.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Reasoned teacher answer

Z: its pages is an integer and syntax is valid, but 46 contradicts approved 64.

Changed-case reasoning: The changed object fails **schema** despite keeping the correct page count. Exactly means additional **field**s are not allowed in this contract.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Structured Output and Schemas investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Validate six **output** cards for required **field**s, **type**s, allowed values, and **evidence** support; explain each rejection.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim

Record	What to capture
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Valid JSON means correct answer: JSON syntax says nothing about **schema** or source truth.

Python treats bool as int, so it is acceptable: The contract explicitly excludes Booleans; an implementation must check accordingly.

A valid report authorises a side effect: Reporting and booking have different permissions.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Structured Output and Schemas questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Independent question

Independent question · withhold the key

A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

Expected independent answer

> **Teacher key · Expected independent answer**
>
> The count value is text instead of an integer. A valid replacement is {"title":"Seeds","count":3}. Valid structure still does not prove factual correctness.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “JSON that looks tidy necessarily satisfies the **schema**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Use {"title":"Cloud","pages":80}. This satisfies the stated **type**s but the page count still needs a reliable edition-specific source.

Reject malformed or missing **field**s explicitly rather than assuming a successful-looking response is valid.

Explanations for the additional workbook examples

W1 Distinguish the layers

Workbook question · W1

Explain what parsing, **schema** checks and source support each establish.

Teacher reasoning · W1

Teacher explanation: Parsing establishes grammatical JSON; **schema** checks enforce **field**s/**type**s/conditional rules; semantic checks match the claims to sources and domain meaning. None alone establishes all three.

W2 Audit types and fields

Workbook question · W2

Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Teacher reasoning · W2

Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

W3 Check a citation-like ID

Workbook question · W3

Sources P,Q support B capacity 12. A **schema**-valid object cites P,Q but gives 99. Which layer rejects it?

Teacher reasoning · W3

Teacher explanation: Semantic/source **validation** rejects unsupported capacity 99. Existing source IDs do not establish entailment.

W4 Validate new objects independently

Workbook question · W4

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

Teacher reasoning · W4

Teacher explanation: X passes parsing, **schema** and source **validation**. Y parses but fails **schema** because capacity is a string. Z parses and satisfies **field type**s but fails semantic support because sources name F, not G.

Assuming these values are encoded as proper JSON, syntax is not the failure.

W5 Keep permissions separate

Workbook question · W5

A fully valid supported reporting object is returned. May the application automatically book the room? Explain.

Teacher reasoning · W5

Teacher explanation: No. The specification is read-only; **validation** of a report does not grant action permission. Booking requires a separately authorised action contract and application controls.

Assess independent transfer

Independent question · withhold the key

Selected case: Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: X passes parsing, **schema** and source **validation**. Y parses but fails **schema** because capacity is a string. Z parses and satisfies **field type**s but fails semantic support because sources name F, not G.

Assuming these values are encoded as proper JSON, syntax is not the failure.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Structured Output and Schemas to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Require answer, source snippet, citation, uncertainty, and next-question **field**s.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 9: Prompt Evaluation. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires X passes all, Y **schema** failure, Z semantic failure. Fresh retry: sources approve H capacity 0.

Object H/0 passes; H/null with supported fails **schema**; H/1 with integer capacity passes **schema** but fails semantic support. Preserve legitimate zero.

Week 13 compares prompt specifications using fixed development cases, observable criteria and reserved evaluation.

Plan the delayed return

Suggested return points in this level: ai-3/9, ai-3/11, ai-3/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Structured Output and Schemas lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Separate JSON parsing, **schema validation** and source support checks

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Why do string "12",boolean true,missing capacity and an extra action **field** violate this contract?

Supported practice answer

Reasoned practice answer: Teacher explanation: Capacity must be an integer for supported and null for needs_review, with the **field** present. A Boolean is explicitly excluded.

Extra **field**s violate the **closed schema**; string values are not silently converted.

Check every learner before discussion: Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: JSON syntax says nothing about **schema** or source truth.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Require a **prediction** and a reason before comparing. Expected result: The changed object fails **schema** despite keeping the correct page count.

Reason: Exactly means additional **field**s are not allowed in this contract.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

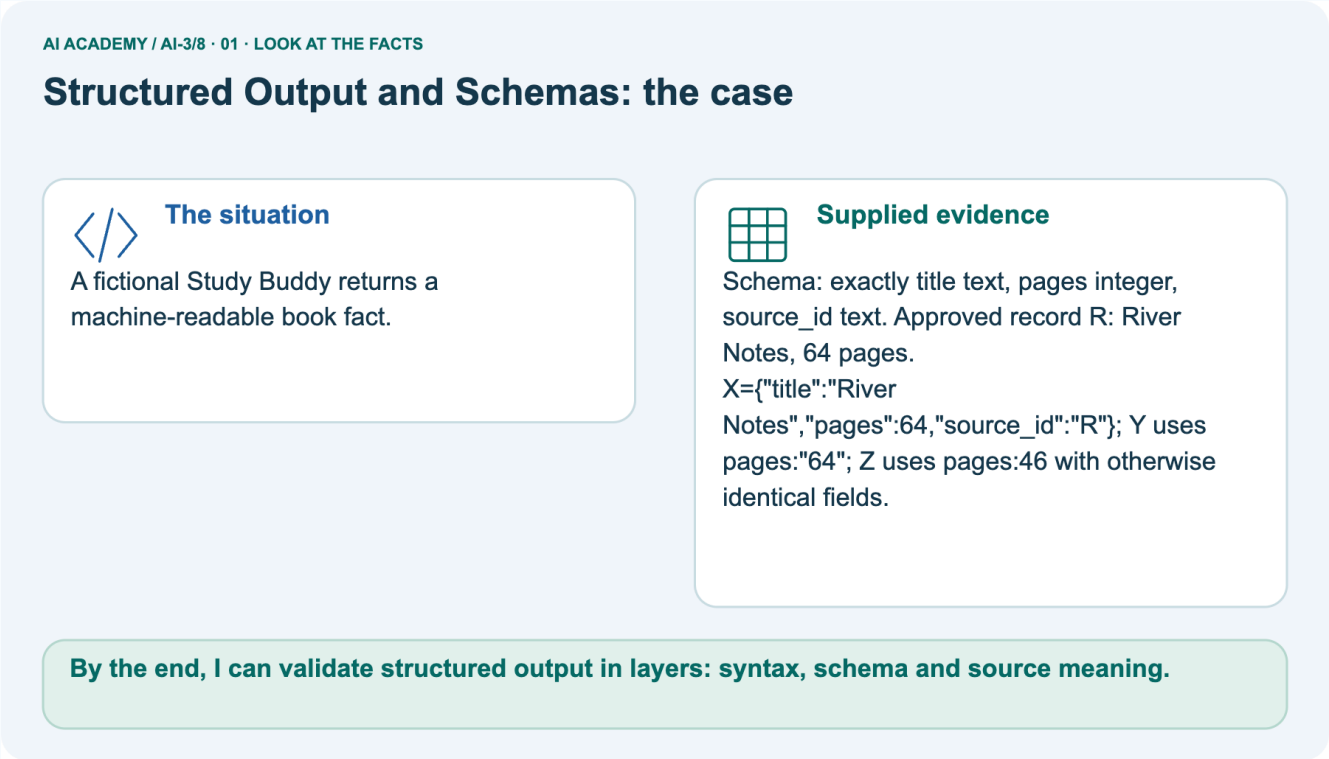
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Structured Output and Schemas graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Structured Output and Schemas: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy returns a machine-readable book fact. Supplied facts: **Schema**: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

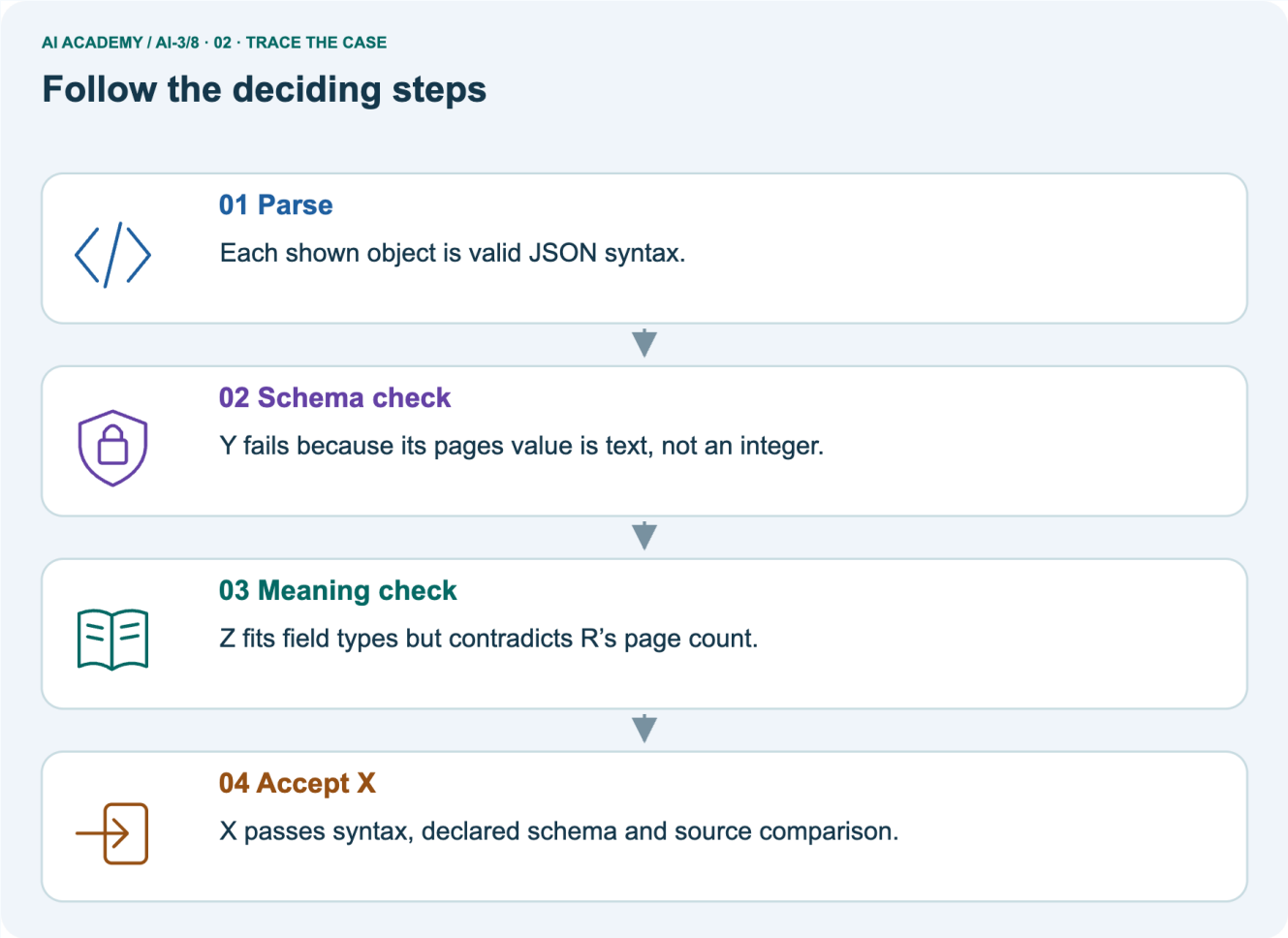
X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

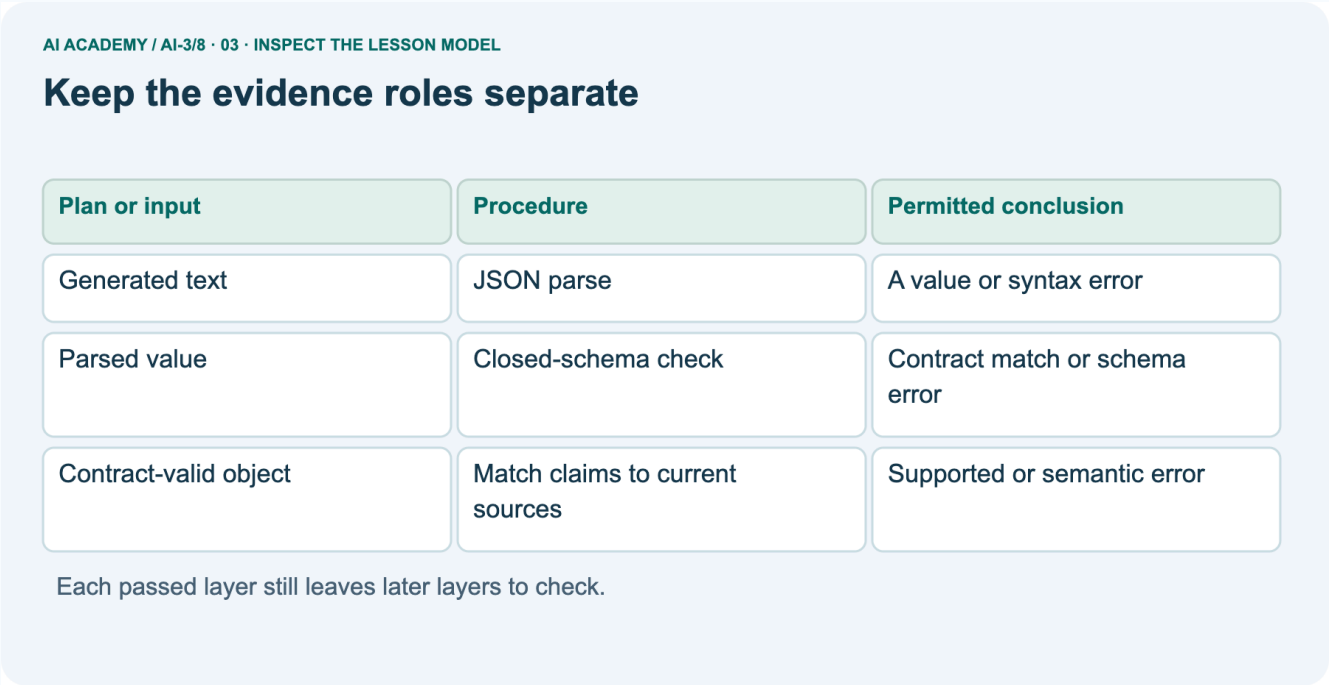
Parse: Each shown object is valid JSON syntax. → **Schema** check: Y fails because its pages value is text, not an integer. → Meaning check: Z fits **field type**s but contradicts R's page count. → Accept X: X passes syntax, declared **schema** and source comparison.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Generated text | JSON parse | A value or syntax error. Parsed value | Closed-**schema** check | Contract match or **schema** error.

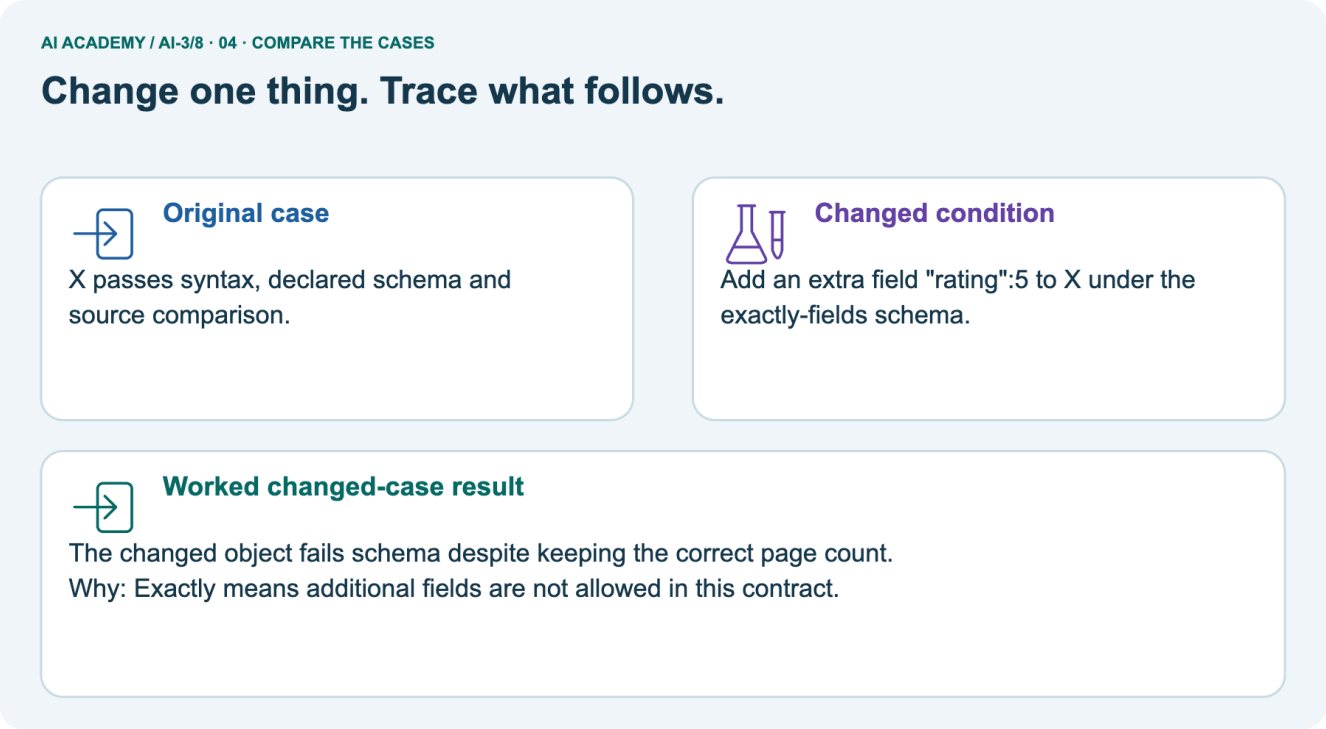
Contract-valid object | Match claims to current sources | Supported or semantic error

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: X passes syntax, declared **schema** and source comparison. Changed condition: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Expected result: The changed object fails **schema** despite keeping the correct page count. Reason: Exactly means additional **field**s are not allowed in this contract.

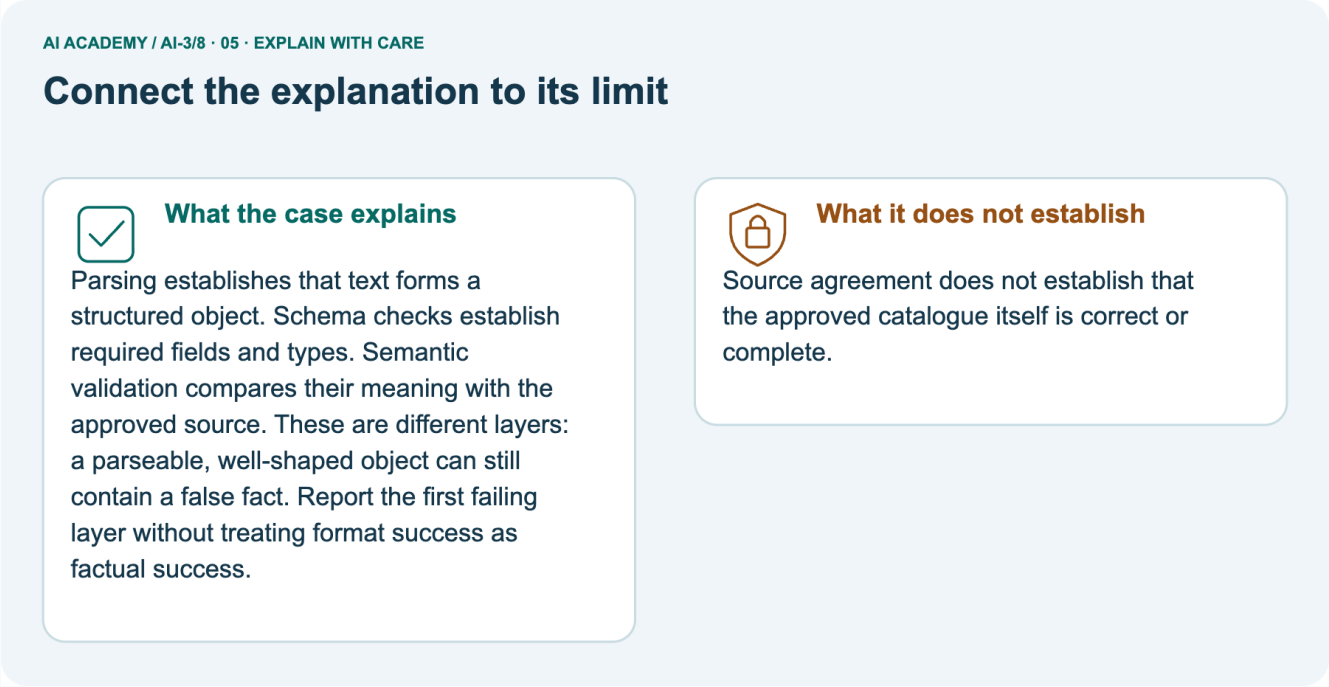
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Exactly means additional **field**s are not allowed in this contract.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Limit: Source agreement does not establish that the approved catalogue itself is correct or complete.

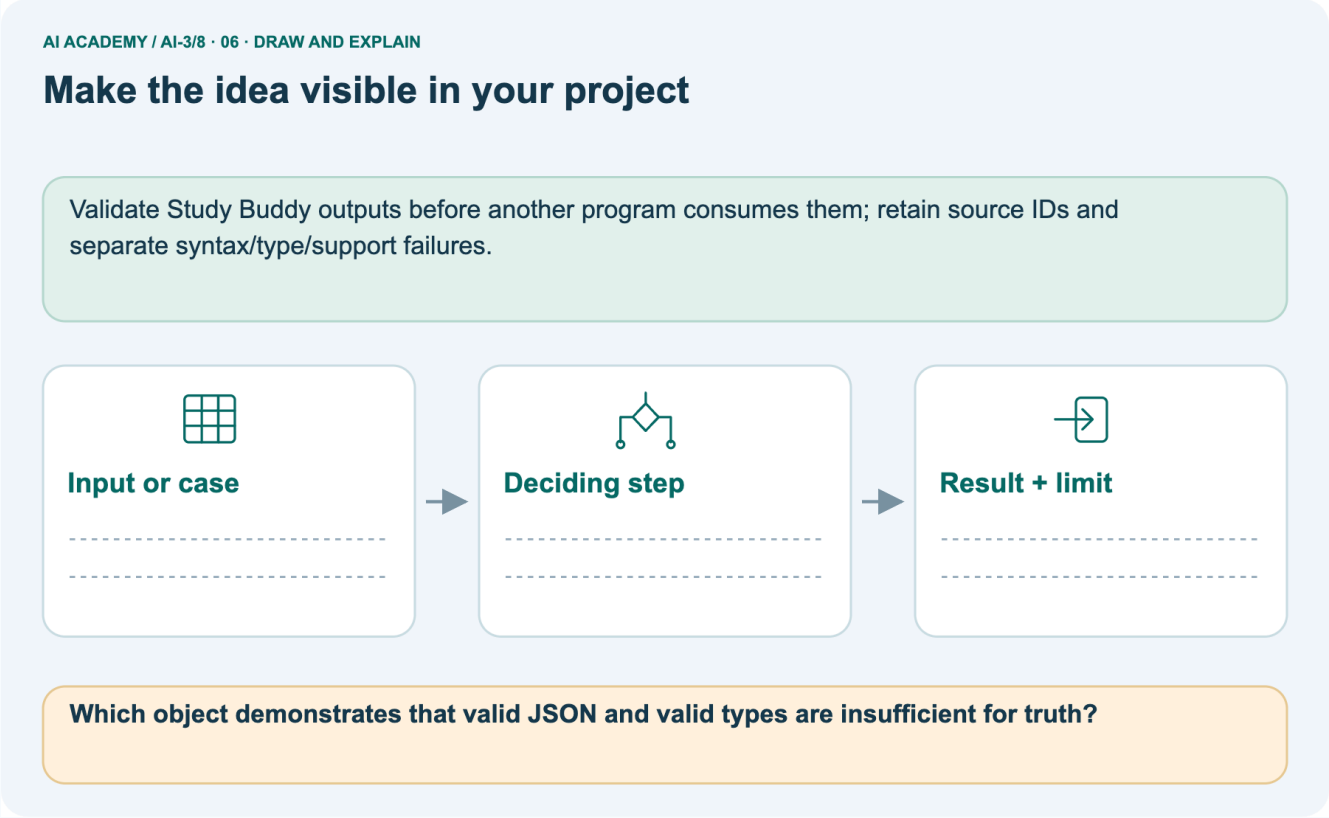
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Source agreement does not establish that the approved catalogue itself is correct or complete.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures. Draw your **input**, deciding step, result and limit.

Supported question: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Z: its pages is an integer and syntax is valid, but 46 contradicts approved 64.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.